

Vansh Goel
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Professional Summary

Computer Science undergraduate with a strong passion for Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Large Language Models (LLMs). Experienced in developing intelligent AI applications using PyTorch, TensorFlow, Scikit-learn, Retrieval-Augmented Generation (RAG), LangChain, and modern deep learning frameworks. Proficient in designing end-to-end machine learning pipelines, feature engineering, model training, evaluation, and hyperparameter optimization. Strong foundation in Python, mathematics, statistics, and collaborative software development using Git, with a focus on building scalable and impactful AI solutions.

Technical Skills

Programming Languages: Python, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transfer Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Deep Learning

LLMs & NLP: Transformers, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Embeddings, Vector Databases

Computer Vision: OpenCV, Image Processing, Image Classification, Image Captioning, Feature Extraction

Data Science & Visualization: NumPy, Pandas, Matplotlib

Frameworks & Tools: FastAPI, Flask, Git, GitHub, Jupyter Notebook

Mathematics: Linear Algebra, Probability, Statistics

Experience

Machine Learning Trainee | Coding Ninjas (Sept 2025 – May 2026)

- Developed supervised machine learning models using Python and Scikit-learn.
- Engineered data preprocessing and feature engineering pipelines.
- Optimized model performance using experimentation and hyperparameter tuning.
- Collaborated using Git and version control best practices.
- Applied testing and debugging techniques to improve model reliability.

Projects

Cat vs Dog Image Classification

Developed a CNN-based image classification model using PyTorch. Applied transfer learning, data augmentation, and hyperparameter tuning. Evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Built a scalable inference pipeline.

Image Captioning System

Developed a multimodal image captioning system by combining CNN-based visual feature extraction with sequence generation techniques. Implemented tokenization, vocabulary creation, and caption generation pipelines. Evaluated using the BLEU score and improved caption quality through optimization.

RAG Assistant Chatbot

Designed and implemented a Retrieval-Augmented Generation (RAG) assistant using LangChain, vector databases, semantic search, and LLMs. Built an end-to-end retrieval

pipeline using embeddings and optimized prompt engineering for contextual responses.

Education

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science Engineering (Expected 2028)